GATE DA 2027

Calculus & Optimization

Topic-wise Question Bank

Weighted by what GATE DA 2024–2026 actually asked

85 practice questions + **6** real previous-year questions • **8** topics

MCQ • MSQ • NAT — the three GATE question types

Answer key at the end — answers only, no worked solutions

Every numerical answer in this book (B^+-tree fan-out, FD closures, SQL counts, z-scores) was recomputed independently in Python before the key was fixed — all 16 machine checks agreed. The 6 previous-year questions carry the official GATE answer key. Practice options are shuffled, so the key is not all "A".

Prepared for Ashutosh • August 2026

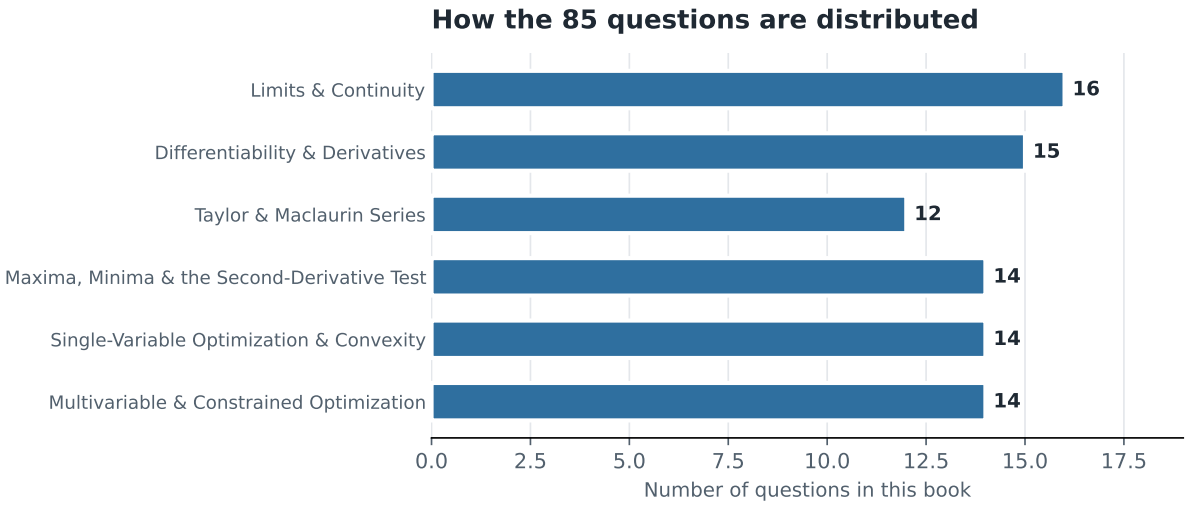
How this book is built

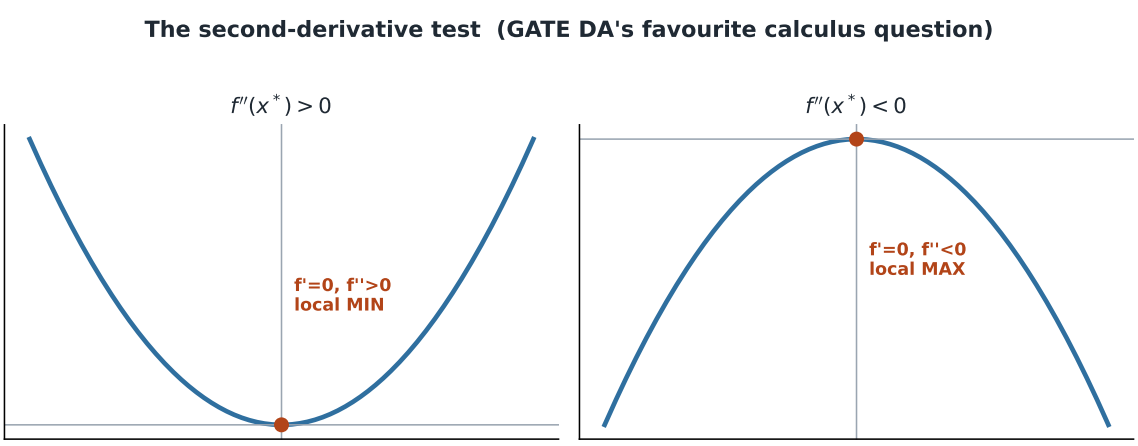
Calculus & Optimization is small but almost free marks — if you drill the handful of patterns GATE actually repeats. It carries roughly **6 to 9 marks** a year, and the same few question types come back: a limit to evaluate, a derivative to compute, and — above all — the **second-derivative test** to classify a stationary point. In 2024 alone the paper asked the second-derivative test (Q.15), a piecewise continuity/differentiability fit (Q.37), a sigmoid derivative (Q.33), critical points of a quartic (Q.50) and a limit (Q.60). This is a subject you can make near-certain marks on with a fortnight of focused practice.

What GATE DA actually asks

1. **Classify a stationary point.** $f'(x^*) = 0$; the *sign of* $f''(x^*)$ decides: positive $\Rightarrow$ minimum, negative $\Rightarrow$ maximum, zero $\Rightarrow$ inconclusive. This single idea is the most repeated calculus question in GATE DA (2024 Q.15, Q.50; 2026 Q.27).
2. **Evaluate a limit** — usually a $\frac{0}{0}$ form cleared by a standard limit, a series expansion, or L'Hôpital (2024 Q.60).
3. **Differentiate something from ML.** The sigmoid, softplus, ReLU, tanh — their derivatives are examined *because* they appear in machine learning (2024 Q.33). Learn them once, use them twice.
4. **Fit a piecewise function** so it is continuous and differentiable — match both the value and the slope at the join (2024 Q.37).

Optimization extends the same second-derivative idea to convex functions (a local minimum is global), to gradient descent (walk opposite the derivative), and to several variables (gradient = 0, Hessian positive definite; Lagrange multipliers for constraints). All of it overlaps directly with Machine Learning, so the time is paid twice.



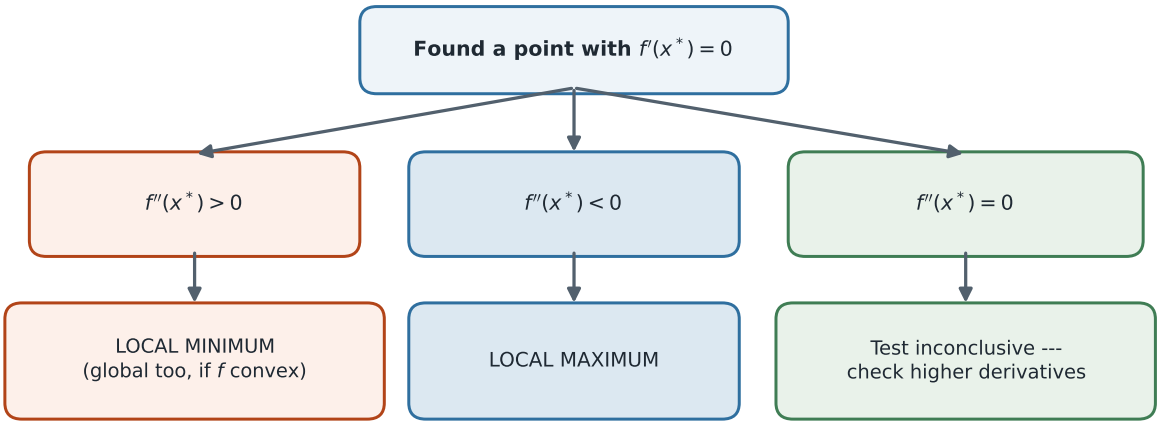


Question types and marking (same as the real paper)

Type	Description	Negative marking
MCQ	four options, exactly one correct	$-1/3$ (1-mark) or $-2/3$ (2-mark)
MSQ	four options, one *or more* correct; no partial credit	none
NAT	numerical answer, entered on a virtual keypad	none

This bank is 61% MCQ, 2% MSQ and 36% NAT.

The decision that answers most calculus questions



Sixteen facts that answer most Calculus & Optimization questions.

- Standard limits: $\frac{\sin x}{x} \to 1$, $\frac{1-\cos x}{x^2} \to \frac{1}{2}$, $\frac{e^x-1}{x} \to 1$, $(1+\frac{1}{x})^x \to e$.
- L'Hôpital applies only to $\frac{0}{0}$ or $\frac{\infty}{\infty}$.
- Continuous $\neq$ differentiable: $|x|$ is continuous but not differentiable at 0; ReLU too.
- Differentiable $\Rightarrow$ continuous (the converse fails).
- Second-derivative test: $f' = 0$ and $f'' > 0 \Rightarrow$ min; $f'' < 0 \Rightarrow$ max; $f'' = 0 \Rightarrow$ inconclusive.
- On a closed interval, extrema occur at critical points *or* endpoints.
- Sigmoid derivative $\sigma' = \sigma(1-\sigma)$; softplus derivative is the sigmoid; $\tanh' = 1 - \tanh^2$.
- Maclaurin: $e^x = \sum \frac{x^n}{n!}$; $\sin x = x - \frac{x^3}{3!} + \cdots$; $\cos x = 1 - \frac{x^2}{2!} + \cdots$; $\frac{1}{1-x} = \sum x^n$.
- Linear approximation: $f(a) + f'(a)(x-a)$; quadratic adds $\frac{1}{2}f''(a)(x-a)^2$.
- Convex $\Leftrightarrow f'' \geq 0$; for convex f every local min is global (strict convex $\Rightarrow$ unique).
- Sum of convex functions is convex; $-f$ convex $\Leftrightarrow f$ concave.
- Gradient descent: $x \leftarrow x - \eta f'(x)$ moves toward smaller f.
- Multivariable min: $\nabla f = 0$ and the Hessian is positive definite; indefinite Hessian $\Rightarrow$ saddle.
- Gradient points in the direction of steepest ascent; $-\nabla f$ is steepest descent.
- Lagrange multipliers optimize f subject to equality constraints ($\nabla f = \lambda \nabla g$).
- A quadratic form $x^\mathsf{T} A x$ is convex $\Leftrightarrow A$ is positive semidefinite.

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Part A — Practice questions by topic

1. Limits & Continuity

16 questions • 8 MCQ, 0 MSQ, 8 NAT • Suggested time: 28 minutes

Q.1 [NAT • 1 Mark • Easy]

The value of $\lim_{x \to 0} \dfrac{\sin x}{x}$ is _____.

Q.2 [NAT • 1 Mark • Moderate]

$\lim_{x \to 0} \dfrac{1 - \cos x}{x^2}$ equals _____ (two decimals).

Q.3 [NAT • 1 Mark • Easy]

$\lim_{x \to 0} \dfrac{e^x - 1}{x}$ equals _____.

Q.4 [MCQ • 1 Mark • Moderate]

$\lim_{x \to \infty} \left(1 + \dfrac{1}{x}\right)^x$ equals

(A) 0 (B) ∞ (C) e (D) 1

Q.5 [NAT • 2 Marks • Hard]

$\lim_{x \to 0} \dfrac{\ln((x^2 + 1)\cos x)}{x^2}$ equals _____ (two decimals). *(GATE DA 2024, Q.60 pattern.)*

Q.6 [MCQ • 1 Mark • Easy]

A function f is continuous at $x = a$ if

(A) $f'(a)$ exists (B) $\lim_{x \to a} f(x) = f(a)$

(C) $f(a) = 0$ (D) f is differentiable at a

Q.7 [MCQ • 1 Mark • Moderate]

The greatest-integer (floor) function $f(x) = \lfloor x \rfloor$ is discontinuous at

(A) no point (B) every real number (C) every integer (D) $x = 0$ only

Q.8 [NAT • 1 Mark • Easy]

$\lim_{x \to 2} \dfrac{x^2 - 4}{x - 2}$ equals _____.

Q.9 [MCQ • 2 Marks • Hard]

For the piecewise function $f(x) = -x$ for $x < -2$, $f(x) = ax^2 + bx + c$ for $x \in [-2, 2]$, and $f(x) = x$ for $x > 2$ to be *continuous and differentiable*, the values are *(GATE DA 2024, Q.37 pattern)*

(A) $a = \frac{1}{4},\ b = 0,\ c = 1$

(B) $a = 0,\ b = 0,\ c = 0$

(C) $a = \frac{1}{2},\ b = 0,\ c = 0$

(D) $a = 1,\ b = 1,\ c = 1$

Q.10 [MCQ • 1 Mark • Moderate]

If $\lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x)$ (both finite), then f has, at $x = a$, a

(A) point of continuity

(B) infinite discontinuity

(C) jump discontinuity

(D) removable discontinuity

Q.11 [NAT • 1 Mark • Easy]

$\lim_{x \to 0} \dfrac{\tan x}{x}$ equals _____.

Q.12 [MCQ • 1 Mark • Moderate]

L'Hôpital's rule may be applied to $\lim_{x \to a} \frac{f(x)}{g(x)}$ when the limit is of the indeterminate form

(A) any form

(B) $\frac{1}{0}$

(C) $\frac{0}{0}$ or $\frac{\infty}{\infty}$

(D) $0 \cdot 1$

Q.13 [NAT • 2 Marks • Hard]

$\lim_{x \to 0^+} x \ln x$ equals _____.

Q.14 [MCQ • 1 Mark • Moderate]

Every differentiable function is continuous, but the converse is false. A standard example of a function continuous everywhere but not differentiable at a point is

(A) $f(x) = x^2$ at $x = 0$

(B) $f(x) = x$ at $x = 0$

(C) $f(x) = e^x$ at $x = 0$

(D) $f(x) = |x|$ at $x = 0$

Q.15 [NAT • 1 Mark • Moderate]

$\lim_{x \to \infty} \dfrac{3x^2 + 2x}{5x^2 - 1}$ equals _____ (two decimals).

Q.16 [MCQ • 1 Mark • Moderate]

The function $f(x) = 1/x$ on $(0, \infty)$ is

(A) continuous at every point of its domain

(B) continuous at $x = 0$

(C) nowhere continuous

(D) discontinuous at $x = 1$

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2. Differentiability & Derivatives

15 questions • 8 MCQ, 0 MSQ, 7 NAT • Suggested time: 27 minutes

Q.1 [MCQ • 1 Mark • Easy]

The derivative of $f(x) = x^n$ is

(A) x^{n-1} (B) $(n-1)x^n$ (C) nx^{n-1} (D) nx^n

Q.2 [MCQ • 1 Mark • Easy]

$\dfrac{d}{dx}\, e^{2x}$ equals

(A) $2x\, e^{2x}$ (B) e^2 (C) e^{2x} (D) $2e^{2x}$

Q.3 [NAT • 2 Marks • Moderate]

The derivative of the sigmoid $\sigma(x) = \dfrac{1}{1+e^{-x}}$ is $\sigma(x)(1-\sigma(x))$. Its value where $\sigma(x) = 0.4$ is _____ (two decimals). *(GATE DA 2024, Q.33 pattern.)*

Q.4 [MCQ • 1 Mark • Easy]

$\dfrac{d}{dx} \ln x$ equals

(A) $\dfrac{1}{x}$ (B) x (C) $-\dfrac{1}{x^2}$ (D) $\ln x$

Q.5 [MCQ • 1 Mark • Moderate]

By the product rule, $\dfrac{d}{dx}[u(x)v(x)]$ equals

(A) $u'v - uv'$ (B) $\dfrac{u'v - uv'}{v^2}$

(C) $u'v'$ (D) $u'v + uv'$

Q.6 [MCQ • 1 Mark • Moderate]

By the quotient rule, $\dfrac{d}{dx}\dfrac{u}{v}$ equals

(A) $\dfrac{u'v - uv'}{v^2}$ (B) $\dfrac{u'}{v'}$

(C) $u'v - uv'$ (D) $\dfrac{u'v + uv'}{v^2}$

Q.7 [NAT • 1 Mark • Easy]

The value of $f'(x)$ at $x = 1$ for $f(x) = x^3 - 3x$ is _____.

Q.8 [NAT • 1 Mark • Easy]

$\dfrac{d}{dx}\sin x$ evaluated at $x = 0$ equals _____.

Q.9 [MCQ • 2 Marks • Hard]

For $f(x) = \ln(1 + e^x)$ (the softplus), $f'(x)$ equals

(A) $\ln(1 + e^x)$

(B) $\dfrac{1}{1 + e^{-x}}$ (the sigmoid)

(C) e^x

(D) $\dfrac{1}{1 + e^x}$

Q.10 [MCQ • 1 Mark • Moderate]

The chain rule gives $\dfrac{d}{dx} f(g(x)) =$

(A) $f'(g(x))\, g'(x)$ (B) $f(g'(x))$ (C) $f'(g(x))$ (D) $f'(x)g'(x)$

Q.11 [NAT • 1 Mark • Moderate]

If $f(x) = x^2 e^x$, then $f'(0)$ equals _____.

Q.12 [NAT • 1 Mark • Moderate]

The second derivative $f''(x)$ of $f(x) = x^3$ at $x = 2$ is _____.

Q.13 [MCQ • 1 Mark • Moderate]

The ReLU function $f(x) = \max(0, x)$ is

(A) not differentiable at $x = 0$ (but differentiable elsewhere)

(B) differentiable everywhere

(C) discontinuous at $x = 0$

(D) nowhere differentiable

Q.14 [NAT • 1 Mark • Moderate]

For $f(x) = \tanh(x)$, the derivative is $1 - \tanh^2(x)$; its value at $x = 0$ is _____.

Q.15 [NAT • 2 Marks • Hard]

The linear approximation of f near $x = a$ is $f(a) + f'(a)(x - a)$. For $f(x) = \sqrt{x}$ near $a = 100$, the estimate of $\sqrt{102}$ (two decimals) is _____.

3. Taylor & Maclaurin Series

12 questions • 8 MCQ, 0 MSQ, 4 NAT • Suggested time: 24 minutes

Q.1 [MCQ • 1 Mark • Easy]

The Taylor series of f about $x = a$ is $\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$. A Maclaurin series is the special case with

(A) $a = \infty$ (B) $a = 1$ (C) $a = f(0)$ (D) $a = 0$

Q.2 [MCQ • 1 Mark • Easy]

The Maclaurin series of e^x is

(A) $x - \frac{x^3}{3!} + \cdots$ (B) $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$

(C) $1 + x^2 + x^4 + \cdots$ (D) $1 - x + \frac{x^2}{2!} - \cdots$

Q.3 [NAT • 1 Mark • Moderate]

The coefficient of x^2 in the Maclaurin series of e^x is _____ (two decimals).

Q.4 [MCQ • 1 Mark • Moderate]

The Maclaurin series of $\sin x$ begins

(A) $1 + x + \frac{x^2}{2!} + \cdots$ (B) $1 - \frac{x^2}{2!} + \cdots$

(C) $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$ (D) $x + \frac{x^3}{3!} + \cdots$

Q.5 [MCQ • 1 Mark • Moderate]

The Maclaurin series of $\cos x$ begins

(A) $1 + \frac{x^2}{2!} + \cdots$ (B) $1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$

(C) $x - \frac{x^3}{3!} + \cdots$ (D) $1 - x + \cdots$

Q.6 [NAT • 2 Marks • Moderate]

Using the first two nonzero terms, $\ln(1 + x) \approx x - \frac{x^2}{2}$. The estimate of $\ln(1.1)$ (two decimals) is _____.

Q.7 [MCQ • 1 Mark • Moderate]

The first-order Taylor (linear) approximation of $f(x) = e^x$ about $x = 0$ is

(A) $1 + x + x^2$ (B) $1 - x$ (C) x (D) $1 + x$

Q.8 [NAT • 2 Marks • Moderate]

The coefficient of x^3 in the Maclaurin series of $\sin x$ is _____ (three decimals).

Q.9 [MCQ • 1 Mark • Moderate]

The Maclaurin series of $\dfrac{1}{1-x}$ (for $|x| < 1$) is

(A) $1 + x + x^2 + x^3 + \cdots$ (B) $1 + 2x + 3x^2 + \cdots$

(C) $x + x^2 + \cdots$ (D) $1 - x + x^2 - \cdots$

Q.10 [MCQ • 1 Mark • Moderate]

The quadratic Taylor approximation of f about a is $f(a) + f'(a)(x-a) + \frac{1}{2}f''(a)(x-a)^2$. For $f(x) = \cos x$ about $a = 0$, this is

(A) $1 - \dfrac{x^2}{2}$ (B) $1 - x$ (C) $x - \dfrac{x^2}{2}$ (D) $1 + \dfrac{x^2}{2}$

Q.11 [MCQ • 2 Marks • Hard]

Taylor's theorem with remainder expresses $f(x)$ as a partial polynomial plus a remainder term that

(A) bounds the approximation error (B) is the first term

(C) equals $f(x)$ (D) is always zero

Q.12 [NAT • 2 Marks • Hard]

Approximating $e^{0.1}$ by $1 + x + \frac{x^2}{2}$ at $x = 0.1$ gives _____ (four decimals).

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4. Maxima, Minima & the Second-Derivative Test

14 questions • 8 MCQ, 2 MSQ, 4 NAT • Suggested time: 25 minutes

Q.1 [MCQ • 1 Mark • Moderate]

For a twice-differentiable $f : \mathbb{R} \to \mathbb{R}$, if $f'(x^*) = 0$ and $f''(x^*) > 0$, then f necessarily has a ___ at x^*. *(GATE DA 2024, Q.15 pattern.)*

(A) global maximum (B) local minimum (C) global minimum (D) local maximum

Q.2 [MCQ • 1 Mark • Easy]

If $f'(x^*) = 0$ and $f''(x^*) < 0$, then x^* is a

(A) saddle/inflection (B) global minimum (C) local minimum (D) local maximum

Q.3 [MCQ • 1 Mark • Easy]

A point where $f'(x) = 0$ is called a

(A) global maximum (B) discontinuity

(C) inflection point only (D) stationary (critical) point

Q.4 [NAT • 1 Mark • Easy]

For $f(x) = x^2 - 4x + 7$, the value of x at which f attains its minimum is _____.

Q.5 [NAT • 1 Mark • Moderate]

For $f(x) = x^2 - 4x + 7$, the minimum value of f is _____.

Q.6 [MSQ • 2 Marks • Hard]

$f(x) = \frac{x^4}{4} - \frac{2x^3}{3} - \frac{3x^2}{2} + 1$ has $f'(x) = x^3 - 2x^2 - 3x = x(x-3)(x+1)$. Which statement(s) is/are TRUE? *(GATE DA 2024, Q.50 pattern.)*

(A) $x = -1$ is a local maximum (B) $x = 0$ is a local maximum

(C) $x = 3$ is a local minimum (D) $x = 0$ is a local minimum

Q.7 [MCQ • 1 Mark • Moderate]

If $f''(x^*) = 0$ and $f'(x^*) = 0$, the second-derivative test is

(A) inconclusive (need higher-order analysis) (B) a guaranteed minimum

(C) a guaranteed maximum (D) a discontinuity

Q.8 [MCQ • 1 Mark • Moderate]

The critical points of $f(x) = x^3 - 3x$ are at $x =$

(A) ± 3 (B) 0 only (C) ± 1 (D) $0, 3$

Q.9 [MCQ • 1 Mark • Moderate]

For $f(x) = x^3 - 3x$, the point $x = 1$ is a

(A) local minimum (B) local maximum (C) inflection point (D) global maximum

Q.10 [MCQ • 1 Mark • Moderate]

On a closed interval $[a, b]$, a continuous function attains its absolute maximum and minimum

(A) only at the midpoint

(B) only at critical points

(C) at a critical point or an endpoint

(D) only at endpoints

Q.11 [MSQ • 2 Marks • Hard]

$f(x) = x^3 - 3x^2 + 2$ on $(-1, 3]$. Which statement(s) is/are correct? *(GATE DA 2026, Q.27 pattern.)*

(A) f has a root at $x = 1$

(B) f has exactly two roots in $[-0.9, 0]$

(C) f has a maximum at $x = 0$ *only*

(D) f has a minimum at $x = 2$

Q.12 [MCQ • 1 Mark • Moderate]

An inflection point of f is where

(A) $f = 0$

(B) the concavity changes sign (typically $f'' = 0$ and changes sign)

(C) f is discontinuous

(D) $f' = 0$

Q.13 [NAT • 1 Mark • Moderate]

The maximum value of $f(x) = 6x - x^2$ is _____.

Q.14 [NAT • 2 Marks • Hard]

For $f(x) = x + \dfrac{1}{x}$ on $x > 0$, the minimum value is _____.

5. Single-Variable Optimization & Convexity

14 questions • 9 MCQ, 0 MSQ, 5 NAT • Suggested time: 25 minutes

Q.1 [MCQ • 1 Mark • Moderate]

A twice-differentiable function f is *convex* on an interval if

(A) $f(x) \geq 0$

(B) $f'(x) \geq 0$

(C) $f''(x) \geq 0$ throughout the interval

(D) $f''(x) \leq 0$ throughout

Q.2 [MCQ • 1 Mark • Moderate]

For a convex function, any local minimum is

(A) also a global minimum

(B) a saddle point

(C) a local maximum

(D) never global

Q.3 [MCQ • 1 Mark • Moderate]

Which of the following functions is convex on all of $\mathbb{R}$?

(A) $f(x) = -x^2$ (B) $f(x) = \sin x$ (C) $f(x) = x^2$ (D) $f(x) = \ln x$

Q.4 [MCQ • 1 Mark • Moderate]

The function $f(x) = e^x$ is

(A) linear

(B) convex on all of $\mathbb{R}$ ($f'' = e^x > 0$)

(C) neither convex nor concave

(D) concave

Q.5 [NAT • 1 Mark • Moderate]

To minimize the cost $C(x) = x^2 - 20x + 150$ (an open box's material cost), the optimal x is _____.

Q.6 [NAT • 2 Marks • Moderate]

A rectangle has a fixed perimeter of 20. Its area is maximized when it is a square of side _____.

Q.7 [NAT • 2 Marks • Moderate]

For that rectangle (perimeter 20), the maximum area is _____.

Q.8 [MCQ • 1 Mark • Moderate]

Gradient descent on a differentiable f uses the update $x \leftarrow x - \eta f'(x)$. It moves x

(A) to $x = 0$

(B) toward increasing f

(C) to a random point

(D) in the direction of decreasing f (opposite the gradient)

Q.9 [NAT • 1 Mark • Moderate]

One gradient-descent step on $f(x) = x^2$ from $x = 5$ with learning rate $\eta = 0.1$ gives $x =$ _____.

Q.10 [MCQ • 1 Mark • Moderate]

A function that is convex and whose derivative is zero at x^* has, at x^*, its

(A) a saddle point (B) global minimum (C) an inflection (D) global maximum

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Q.11 [MCQ • 1 Mark • Moderate]

The sum of two convex functions is

(A) concave (B) convex (C) neither (D) always linear

Q.12 [MCQ • 1 Mark • Easy]

A function f is *concave* if $-f$ is

(A) convex (B) increasing (C) concave (D) linear

Q.13 [NAT • 2 Marks • Hard]

To minimize $f(x) = x^2 + \dfrac{16}{x}$ for $x > 0$, the optimal x (setting $f' = 0$) is _____.

Q.14 [MCQ • 1 Mark • Moderate]

For a strictly convex function, the global minimizer is

(A) at infinity (B) never unique (C) always at $x = 0$ (D) unique

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6. Multivariable & Constrained Optimization

14 questions • 11 MCQ, 0 MSQ, 3 NAT • Suggested time: 27 minutes

Q.1 [MCQ • 1 Mark • Moderate]

For $f(x, y)$, the gradient ∇f is the vector

(A) a scalar

(B) $\left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right)$

(C) $\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}$

(D) $\frac{\partial^2 f}{\partial x\, \partial y}$

Q.2 [MCQ • 1 Mark • Moderate]

At a local minimum of a differentiable $f(x, y)$, the gradient ∇f equals

(A) a unit vector (B) the zero vector (C) the Hessian (D) 1

Q.3 [MCQ • 1 Mark • Moderate]

For $f(x, y) = x^2 + y^2$, the gradient at $(1, 2)$ is

(A) $(1, 2)$ (B) $(4, 2)$ (C) $(2, 4)$ (D) $(2, 2)$

Q.4 [MCQ • 1 Mark • Moderate]

The Hessian of $f(x, y)$ is the matrix of

(A) gradients (B) first-order partial derivatives

(C) eigenvalues (D) second-order partial derivatives

Q.5 [MCQ • 1 Mark • Moderate]

A stationary point of $f(x, y)$ is a local minimum if the Hessian there is

(A) singular (B) negative definite (C) positive definite (D) indefinite

Q.6 [MCQ • 1 Mark • Moderate]

If the Hessian at a stationary point is *indefinite* (both positive and negative eigenvalues), the point is a

(A) local maximum (B) saddle point (C) local minimum (D) global minimum

Q.7 [MCQ • 1 Mark • Moderate]

For $f(x, y) = x^2 + 3y^2$, the minimum over all (x, y) is attained at

(A) $(1, 1)$ (B) $(0, 1)$ (C) $(0, 0)$ (D) $(1, 0)$

Q.8 [MCQ • 1 Mark • Moderate]

The method of *Lagrange multipliers* is used to

(A) optimize a function subject to equality constraints
(B) invert the Hessian
(C) compute a limit
(D) differentiate a product

Q.9 [NAT • 2 Marks • Hard]

To minimize $f(x, y) = x^2 + y^2$ subject to $x + y = 1$, the Lagrange conditions give the minimizer $x = y =$ _____ (two decimals).

Q.10 [NAT • 2 Marks • Hard]

For that problem ($\min x^2 + y^2$ s.t. $x + y = 1$), the minimum value is _____ (two decimals).

Q.11 [MCQ • 2 Marks • Hard]

A quadratic form $x^{\mathsf{T}}Ax$ is convex if and only if the (symmetric) matrix A is

(A) skew-symmetric
(B) positive semidefinite
(C) invertible
(D) orthogonal

Q.12 [MCQ • 1 Mark • Moderate]

For $f(x, y) = (x - 2)^2 + (y + 3)^2$, the global minimizer is

(A) $(0, 0)$ (B) $(-2, 3)$ (C) $(2, 3)$ (D) $(2, -3)$

Q.13 [MCQ • 1 Mark • Moderate]

The directional derivative of f is maximized in the direction of

(A) $-\nabla f$
(B) any direction equally
(C) the Hessian's eigenvector
(D) the gradient ∇f

Q.14 [NAT • 2 Marks • Moderate]

For $f(x, y) = x^2 + y^2$ the Hessian is $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$; its eigenvalues are both _____, confirming a minimum.

Part B — Previous-year questions (with official keys)

These are **genuine questions from the official GATE DA papers** in your archive. Each carries the **official GATE answer key** and its options are kept in the original paper order (not shuffled). Every question is tagged with its exact year and number. Grouped under the same topics as Part A.

1. Limits & Continuity

Q.1 PYQ [NAT • 2 Marks • Hard]

Evaluate the following limit: $\lim_{x \to 0} \dfrac{\ln\left((x^2+1)\cos x\right)}{x^2}$. The value is _____.

Source: GATE DA 2024, Q.60 (2 marks) — official paper & official answer key

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Q.2 PYQ [MCQ • 2 Marks • Hard]

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = -x$ if $x < -2$; $f(x) = ax^2 + bx + c$ if $x \in [-2, 2]$; and $f(x) = x$ if $x > 2$. Which one of the following gives values of a, b, c that make f continuous and differentiable?

(A) $a = \frac{1}{4},\ b = 0,\ c = 1$ (B) $a = \frac{1}{2},\ b = 0,\ c = 0$

(C) $a = 0,\ b = 0,\ c = 0$ (D) $a = 1,\ b = 0,\ c = 1$

Source: GATE DA 2024, Q.37 (2 marks) — official paper & official answer key

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2. Differentiability & Derivatives

Q.1 PYQ [NAT • 1 Mark • Moderate]

Let $f : \mathbb{R} \to \mathbb{R}$ be $f(x) = \dfrac{1}{1 + e^{-x}}$. The value of the derivative of f at the x where $f(x) = 0.4$ is _____ (rounded to two decimal places).

Source: GATE DA 2024, Q.33 (1 mark) — official paper & official answer key

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4. Maxima, Minima & the Second-Derivative Test

Q.1 PYQ [MCQ • 1 Mark • Moderate]

For any twice differentiable function $f : \mathbb{R} \to \mathbb{R}$, if at some $x^* \in \mathbb{R}$, $f'(x^*) = 0$ and $f''(x^*) > 0$, then f necessarily has a _____ at $x = x^*$.

(A) local minimum (B) global minimum (C) local maximum (D) global maximum

Source: GATE DA 2024, Q.15 (1 mark) — official paper & official answer key

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Q.2 PYQ [MSQ • 2 Marks • Hard]

Consider $f : \mathbb{R} \to \mathbb{R}$ with $f(x) = \dfrac{x^4}{4} - \dfrac{2x^3}{3} - \dfrac{3x^2}{2} + 1$. Which of the following statements is/are TRUE?

(A) $x = 0$ is a local maximum of f (B) $x = 3$ is a local minimum of f

(C) $x = -1$ is a local maximum of f (D) $x = 0$ is a local minimum of f

Source: GATE DA 2024, Q.50 (2 marks) — official paper & official answer key

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Q.3 **PYQ** [MSQ • 2 Marks • Hard]

Let $f(x) = x^3 - 3x^2 + 2$ be a function defined on $(-1, 3]$. Which of the following statements is/are correct?

(A) $f(x)$ has exactly two roots in $[-0.9, 0]$ (B) $f(x)$ has a minimum at 2 only

(C) $f(x)$ has maximum at 0 only (D) $f(x)$ has a root at 1

Source: GATE DA 2026, Q.27 (1 mark) — official paper & official answer key

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Answer Key

Answers only. A key with several letters (e.g. **AB**) is an MSQ, where every listed option must be selected and no other. For NAT questions any value within the usual GATE tolerance of the figure shown is accepted.

Part A — Practice

1. Limits & Continuity

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	1	**2**	0.5	**3**	1	**4**	C	**5**	0.5
6	B	**7**	C	**8**	4	**9**	A	**10**	C
11	1	**12**	C	**13**	0	**14**	D	**15**	0.6
16	A								

2. Differentiability & Derivatives

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	**2**	D	**3**	0.24	**4**	A	**5**	D
6	A	**7**	0	**8**	1	**9**	B	**10**	A
11	0	**12**	12	**13**	A	**14**	1	**15**	10.10

3. Taylor & Maclaurin Series

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	**2**	B	**3**	0.5	**4**	C	**5**	B
6	0.10	**7**	D	**8**	-0.167	**9**	A	**10**	A
11	A	**12**	1.1050						

4. Maxima, Minima & the Second-Derivative Test

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	**2**	D	**3**	D	**4**	2	**5**	3
6	BC	**7**	A	**8**	C	**9**	A	**10**	C
11	AD	**12**	B	**13**	9	**14**	2		

5. Single-Variable Optimization & Convexity

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	**2**	A	**3**	C	**4**	B	**5**	10
6	5	**7**	25	**8**	D	**9**	4	**10**	B
11	B	**12**	A	**13**	2	**14**	D		

6. Multivariable & Constrained Optimization

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	**2**	B	**3**	C	**4**	D	**5**	C
6	B	**7**	C	**8**	A	**9**	0.5	**10**	0.5
11	B	**12**	D	**13**	D	**14**	2		

Part B — Previous-year questions (official keys)

Source	Q	Official answer
GATE DA 2024, Q.60 (2 marks)	C1.1	**0.5**
GATE DA 2024, Q.37 (2 marks)	C1.2	**A**
GATE DA 2024, Q.33 (1 mark)	C2.1	**0.24**
GATE DA 2024, Q.15 (1 mark)	C4.1	**A**
GATE DA 2024, Q.50 (2 marks)	C4.2	**AB**
GATE DA 2026, Q.27 (1 mark)	C4.3	**BD**

Score sheet

Anything below 70% on a topic means the concept, not the practice, is the problem — go back and revise before re-attempting.

Topic	Questions	Correct	Time taken
Limits & Continuity	16		
Differentiability & Derivatives	15		
Taylor & Maclaurin Series	12		
Maxima, Minima & the Second-Derivative Test	14		
Single-Variable Optimization & Convexity	14		
Multivariable & Constrained Optimization	14		
Total	**85**		